Classification of trajectories of dynamic systems using physically-informed neural networks

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Abstract

The paper proposes a method for multivariate time series classification that uses prior information about the physical nature of the measured signals. In contrast to models that rely on assumptions about the statistical properties of trajectories only, the proposed model describes the mechanical system that generates a trajectory . We propose to classify not the trajectories themselves, but the dynamical systems given by the Lagrangians that correspond to these trajectories. The Lagrangian is reconstructed from measurements by a physics-informed Lagrangian neural network . We introduce a seminorm on the space of Lagrangians , show that the induced quotient identifies exactly those Lagrangians that generate the same accelerations, which is the classical gauge freedom of the Lagrangian, and use the induced metric for classification. Sampling this seminorm on a fixed set of probe points in the phase space yields a fixed-length vector representation of a trajectory, so that any vector classifier is applicable. In the computational experiment on the PAMAP2 human activity dataset the proposed representation separates the activity classes with the accuracy **0.946** for the Gaussian process classifier. We also discuss the main practical limitation of the approach: the Lagrangian of every trajectory is currently recovered by a separate optimization problem.

Keywords: *multivariate time series classification; physical system; Lagrangian; Lagrangian neural network; compactness hypothesis*

1 Introduction

The physics-informed neural networks (PINNs) combine the advantages of statistical and mathematical models. The first ones fit the measured data while the second ones take a physical description of the modeled system into account. The PINNs model conservation laws using data as the basis for predictions. Their architectures are analyzed in the reviews [1, 2] . PINNs model principles of Hamiltonian [3], Lagrangian dynamics [4], including Hamiltonian mechanics with infinite-dimensional measurement space [5] from quantum physics , and constrained Hamiltonian systems treated by the Dirac theory of constraints [6] . The Lagrangian networks (LNN) also adapt to model microsystems using graph networks [7]. The modifications of LNN model complex macro-systems such as fluids with the multi-particle approximation [8]. Also, PINNs allow dimension reduction using a special operator [9]. PINNs and LNNs also reconstruct sound [10] taking into account wave equations.

The paper [11] introduces Hamiltonian neural networks to adapt a nonlinear variation of a parametrized system and determine the parameter of nonlinearity. This parameter is responsible for the chaotic nature of the system, acting as a measure of chaos and separating the chaotic system from the original system. The disadvantage of this classification is that one needs to know in advance the law by which these systems are separated. According to [12] Neural Ordinary Differential Equations (NODE) could be unstable due to changes in the initial data, and the error grows exponentially in time. The architecture of CH-NODE reduces these shortcomings but in general, the classification problem for systems with Lagrangian dynamic remains unexplored.

Mostly convolutional neural networks are used for multivariate classification, under the assumption that there are statistical relationships between the neighbouring points of a time series [13]. These methods perform well in the tasks where such relationships are predominant. For physical systems, however, they discard the available prior information, and we propose to use the knowledge about the physical constraints of the system; for the Lagrangian neural networks used here this knowledge is the law of conservation of energy.

This paper solves the problem of dynamic system trajectory classification. It proposes to use Lagrangian neural networks (LNN) for low-dimensional trajectories. Each trajectory refers to the Lagrangian of the system parameterized by the LNN. The issue of mapping the Lagrangian of the system along the trajectory is analyzed in [4].

The dynamical system's state and its trajectory are constructed with multivariate time series [13]. Deep learning methods, developed for the univariate time series, like AlexNet, ResNet, InceptionTime [14] , TapNet [13, 15] need architecture updates for the data acquired from several sensors [16] including three-axis accelerometer and gyroscope [17] and updates for the knowledge about the physical description of the dynamic system. The strongest general-purpose approach in this field is the family of methods based on randomized convolutional kernels, ROCKET [18] and its faster and more accurate successors MiniRocket [19] and MultiRocket [20]. According to the recent large-scale comparison [21], MultiRocket combined with Hydra and the HIVE-COTE 2.0 ensemble are the most accurate methods over the archive of benchmark problems, yet no single method dominates on all datasets . All these methods treat a time series as a purely statistical object: they extract discriminative shapes and frequencies from the measurements, but they do not use any prior knowledge about the mechanics of the system that produced the measurements. The present paper studies the complementary option, when such prior knowledge is available .

The Lagrangian network reconstructs system trajectories for which the conservation of energy and the Euler-Lagrange equations hold. The LNN uses an independent set of coordinates to specify the state of the system. In the Lagrangian formalism the dynamic system is defined by the Lagrangian function , and the generalized coordinates specify the state of the system. The minimum number of independent variables , the generalized coordinates, necessary to describe the state of the mechanical system is chosen ; denote this number by r . The Lagrangian of the system is assumed to be a twice continuously differentiable and square-integrable function , so that its Euler-Lagrange equation has a unique solution. The dynamic system is described by the Euler-Lagrange equation,

$$\frac{\partial L}{\partial \mathbf{q}} - \frac{d}{dt}\frac{\partial L}{\partial \dot{\mathbf{q}}} = \mathbf{0}. \tag{1}$$

Equation (1) is written for a closed system, so that the generalized non-conservative forces are absent and the right-hand side is the zero vector of $\mathbb{R}^r$. It is a system of r second-order differential equations. It is resolved with respect to the accelerations $\ddot{\mathbf{q}}$ if and only if the matrix of the second derivatives of L with respect to the generalized velocities is non-degenerate,

$$\det \mathbf{H}(L) \neq 0, \quad \text{where} \quad \mathbf{H}(L) = \left[\frac{\partial^2 L}{\partial \dot{q}_j \partial \dot{q}_k}\right]_{j,k=1}^{r} \in \mathbb{R}^{r\times r}. \tag{2}$$

The matrix $\mathbf{H}(L)$ is the Hessian of the Lagrangian with respect to the velocities. For a mechanical system with the kinetic energy $T = \frac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}}\mathbf{M}(\mathbf{q})\dot{\mathbf{q}}$ it coincides with the mass matrix $\mathbf{M}(\mathbf{q})$, and condition (2) states that the masses are non-zero, so that the Cauchy problem for the trajectory has a unique solution. The equivalent physical definition of the Lagrangian is the difference between the kinetic T and the potential V energy,

$$L(\mathbf{q}, \dot{\mathbf{q}}) = T(\mathbf{q}, \dot{\mathbf{q}}) - V(\mathbf{q}), \tag{3}$$

where $\mathbf{q}$ is the vector of the generalized coordinates of the system, $\dot{\mathbf{q}}$ is the vector of the generalized velocities, and L is the Lagrangian of the system. The construction of the vector $\mathbf{q}_t$ from the readings of several sensors at the time-tick t is shown in Figure 1.

For Lagrangians of different motions of a physical system, we assume the compactness hypothesis: the Lagrangians which correspond to different motions lie in different compact and separable subsets. We label these subsets as the classes in the space of

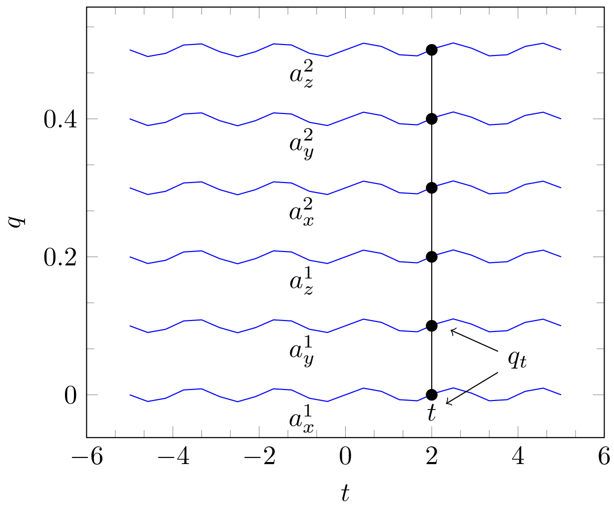


Figure 1: Construction of the state of the dynamic system from the multivariate time series. The readings a_x^k, a_y^k, a_z^k of the k-th sensor at the time-tick t are stacked into one vector $\mathbf{q}_t$ of the generalized coordinates

trajectory models. We assume the parameters of these models are also separable for different classes.

The computational experiment uses the PAMAP2 Physical Activity Monitoring dataset [22]. It contains recordings of nine subjects who performed a fixed protocol of daily physical activities. Each subject wore three inertial measurement units, placed on the wrist of the dominant arm, on the chest, and on the ankle of the dominant side; each unit contains a three-axis accelerometer, a three-axis gyroscope and a three-axis magnetometer sampled at 100 Hz. The protocol part of the dataset, which is used here, contains 12 activities: lying, sitting, standing, ironing, vacuum cleaning, ascending stairs, descending stairs, walking, Nordic walking, cycling, running, and rope jumping; each activity lasts 2–4 minutes. The detailed description of the subset of activities used in the experiment is given in Section 7. Since the dataset provides measurements from several parts of one moving body, it gives an opportunity to evaluate the algorithm on a complex dynamic system. Therefore we stack the measurements of all the sensors used in the experiment into a single vector of generalized coordinates .

2 Formulation of the regression problem for a time series and Lagrangian neural networks

Assume that a Lagrangian L corresponds to a given trajectory of the dynamical system . The Lagrangian defines the dynamical system for which the trajectory is constructed: substituting L into the Euler-Lagrange equation (1) yields a function that relates the coordinates and the velocities of a point of the phase space to its acceleration. Solving this system of ordinary differential equations with the initial condition, which is the initial point of the trajectory, recovers the trajectory completely. We therefore identify each trajectory with a function L from a space equipped with the L_2 -norm, and identify the trajectory with the dynamical system given by L, which is then fed to the input of the classifier. The problem of recovering L from the measurements is formulated as a regression problem.

Denote by X the trajectory of dimension r and length T ,

$$X = [\mathbf{x}_1, \dots, \mathbf{x}_T], \quad \text{where} \quad \mathbf{x}_t = (\mathbf{q}_t, \dot{\mathbf{q}}_t) \in \mathbb{R}^{2r}, \tag{4}$$

$\mathbf{q}_t \in \mathbb{R}^r$ is the vector of generalized coordinates and $\dot{\mathbf{q}}_t \in \mathbb{R}^r$ is the vector of generalized velocities at the t -th time-tick.

There is given a sample set

$$D = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^{T}, \tag{5}$$

where $\mathbf{y}_t = \ddot{\mathbf{q}}_t \in \mathbb{R}^r$ is the acceleration at the t -th time-tick. One has to find a function that predicts the acceleration at an arbitrary point of the phase space,

$$\hat{\mathbf{y}} = \mathbf{f}(\mathbf{x} \mid D), \quad \text{where} \quad \hat{Y} = \{\hat{\mathbf{y}}_t = \hat{\ddot{\mathbf{q}}}_t\}_{t=1}^{T} \tag{6}$$

is the predicted dynamics of the system. The function $\mathbf{f} = \mathbf{f}(\mathbf{x} \mid \mathbf{w})$ with parameters $\mathbf{w}$ is selected from the family of Lagrangian neural networks.

The loss function of the predicted acceleration $\hat{\mathbf{y}}_t$ given the points $\mathbf{x}_t = (\mathbf{q}_t, \dot{\mathbf{q}}_t)$ of the trajectory X is

$$\mathcal{L}(\mathbf{w}) = \mathcal{L}(\mathbf{w} \mid D) = \frac{1}{T} \sum_{t=1}^{T} \|\hat{\mathbf{y}}_t - \mathbf{y}_t\|_2^2. \tag{7}$$

To fit the parameters of the neural network one has to solve the optimization problem

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathbb{W}} \big(\mathcal{L}(\mathbf{w})\big). \tag{8}$$

The algorithm of modeling the dynamic system with Lagrangian is:

1. Find analytical expressions for the kinetic T and the potential V energies.
2. Get the Lagrangian $L = T - V$ in (3).
3. Apply the Euler-Lagrange constraint (1).
4. Solve the resulting system of differential equations.

This algorithm is not applicable when the analytical expressions for T and V are unknown. A straightforward alternative is to parametrize the map from the state to the acceleration directly by a neural network,

$$\hat{\mathbf{y}} = \mathbf{f}(\mathbf{x} \mid \mathbf{w}), \quad \text{where} \quad \mathbf{w} \text{ are the parameters of the neural network.} \tag{9}$$

Such a model reproduces the observed accelerations but does not respect the Lagrangian structure of the system, so its predictions are not guaranteed to conserve energy. To add the prior knowledge of the physics of the system we parametrize the Lagrangian instead of the accelerations,

$$\hat{L} = f(\mathbf{x} \mid \mathbf{w}), \quad \text{where} \quad \hat{L} \text{ is the estimate of the Lagrangian at the point } \mathbf{x}. \tag{10}$$

The key idea is to fit the Lagrangian L by a neural network f : one has to obtain the expression of the Euler-Lagrange constraint and back propagate the error through this constraint ,

$$\ddot{\mathbf{q}} = (\nabla_{\dot{\mathbf{q}}\dot{\mathbf{q}}} L)^{-1} \left[\nabla_{\mathbf{q}} L - (\nabla_{\dot{\mathbf{q}}\mathbf{q}} L)\, \dot{\mathbf{q}}\right]. \tag{11}$$

The diagram of the proposed neural network is

$$\mathbf{x}_i = (\mathbf{q}_i, \dot{\mathbf{q}}_i) \xrightarrow{\hat{L}\colon (\mathbf{x}|\mathbf{w}) \to L_{\mathbf{x}}} L_{\mathbf{x}} \xrightarrow{\nabla L} \frac{d}{dt}\frac{\partial L}{\partial \dot{\mathbf{q}}} = \frac{\partial L}{\partial \mathbf{q}} \xrightarrow{\ddot{\mathbf{q}}_i = g(\mathbf{q}_i, \dot{\mathbf{q}}_i)} \mathbf{y}_i = \ddot{\mathbf{q}}_i\ . \tag{12}$$

The neural network $\hat{L}\colon (\mathbf{x} \mid \mathbf{w}) \to L_{\mathbf{x}}$ from (12) maps the $2r$-dimensional state to a scalar. It is a fully-connected network with the Softplus activation function; the exact architecture is given in Section 7. The activation function must be at least twice continuously differentiable, since the second derivatives of the output with respect to the input enter the model (11); this is why the piecewise-linear ReLU is not applicable here and Softplus is used. Given the estimate $\hat{L}$ of the Lagrangian L, one differentiates $\hat{L}$ by automatic differentiation and obtains $\nabla_{\mathbf{q}}\hat{L}$, $\nabla_{\dot{\mathbf{q}}\dot{\mathbf{q}}}\hat{L}$, and $\nabla_{\dot{\mathbf{q}}\mathbf{q}}\hat{L}$ at the point $\mathbf{x}$. To obtain the acceleration $\hat{\mathbf{y}}$, the values of these derivatives are substituted into the Euler-Lagrange equation (1) written in the form (11). The gradient of the loss (7) is then propagated back through the whole computational graph, including the differentiation operators, to the parameters $\mathbf{w}$. For the given coordinates $(\mathbf{q}, \dot{\mathbf{q}})$ we thus have a model with the prior knowledge of energy conservation which recovers the dynamics $(\dot{\mathbf{q}}, \ddot{\mathbf{q}})$.

3 Approximation of the norm in the Lagrangian space

We assume that the sets of the Lagrangians which correspond to one class of trajectories are compact and pairwise separable . Denote by $\mathbb{L}$ the functional Lagrangian space ,

$$\mathbb{L} = \{L\colon \Omega \to \mathbb{R}\}, \quad \Omega \subset \mathbb{R}^{2r},$$

the set of twice continuously differentiable square-integrable functions on a compact domain Ω of the phase space, where $\mathbf{x} = (\mathbf{q}, \dot{\mathbf{q}}) \in \Omega$ is the vector of generalized coordinates and velocities and $L(\mathbf{x}) \in \mathbb{R}$ is the value of the Lagrangian of the dynamic system. The parametric family of functions (10) approximates the Lagrangian,

$$\mathbb{L}_\epsilon = \{\hat{L}\colon (\mathbf{x} \mid \mathbf{w}) \to \hat{L}(\mathbf{x}) \mid \mathbf{x} \in \Omega,\ \hat{L}(\mathbf{x}) \in \mathbb{R},\ \mathbf{w} \in \mathbb{W}\}, \quad \mathbb{L}_\epsilon \subset \mathbb{L},$$

where $\mathbf{w}$ is the vector of parameters of the approximation model and ϵ is the approximation accuracy.

To construct the seminorm we write out the system of equations from which the accelerations are calculated . The Lagrangian formalism models the dynamic system in the coordinates $\mathbf{x}_t = (\mathbf{q}_t, \dot{\mathbf{q}}_t)$. The system starts in the state $\mathbf{x}_{t_0}$ and ends in the state $\mathbf{x}_{t_1}$. Define the action

$$S = \int_{t_0}^{t_1} L\, dt.$$

The action is a functional on the paths from $\mathbf{x}_{t_0}$ to $\mathbf{x}_{t_1}$ in the time interval $[t_0, t_1]$, and the physical path is a stationary point of S. The stationarity condition is the Euler-Lagrange equation (1) , which defines the dynamics of the system . The accelerations $\ddot{\mathbf{q}}$ are obtained from (1) by the chain rule. Expand the total time derivative through the partial derivatives with respect to the coordinates:

$$\left(\frac{\partial}{\partial \mathbf{q}} \frac{\partial L}{\partial \dot{\mathbf{q}}}\right) \frac{d\mathbf{q}}{dt} + \left(\frac{\partial}{\partial \dot{\mathbf{q}}} \frac{\partial L}{\partial \dot{\mathbf{q}}}\right) \frac{d\dot{\mathbf{q}}}{dt} = \frac{\partial L}{\partial \mathbf{q}}.$$

Rewrite the time derivatives as the velocities and the accelerations:

$$\frac{\partial}{\partial \mathbf{q}} \frac{\partial L}{\partial \dot{\mathbf{q}}} \dot{\mathbf{q}} + \frac{\partial}{\partial \dot{\mathbf{q}}} \frac{\partial L}{\partial \dot{\mathbf{q}}} \ddot{\mathbf{q}} = \frac{\partial L}{\partial \mathbf{q}}.$$

Move the summands so that only the unknown ones remain on the left-hand side:

$$\frac{\partial}{\partial \dot{\mathbf{q}}} \frac{\partial L}{\partial \dot{\mathbf{q}}} \ddot{\mathbf{q}} = \frac{\partial L}{\partial \mathbf{q}} - \frac{\partial}{\partial \mathbf{q}} \frac{\partial L}{\partial \dot{\mathbf{q}}} \dot{\mathbf{q}}.$$

Solve the obtained system of linear equations with respect to the unknown accelerations :

$$\ddot{\mathbf{q}} = \left(\frac{\partial}{\partial \dot{\mathbf{q}}} \frac{\partial L}{\partial \dot{\mathbf{q}}}\right)^{-1} \left[\frac{\partial L}{\partial \mathbf{q}} - \frac{\partial}{\partial \mathbf{q}} \frac{\partial L}{\partial \dot{\mathbf{q}}} \dot{\mathbf{q}}\right].$$

Rewrite the partial derivatives with the nabla-notation:

$$\ddot{\mathbf{q}} = (\nabla_{\dot{\mathbf{q}}\dot{\mathbf{q}}} L)^{-1} \left[\nabla_{\mathbf{q}} L - (\nabla_{\dot{\mathbf{q}}\mathbf{q}} L)\, \dot{\mathbf{q}}\right], \quad \text{where the Hessian} \quad \mathbf{H}(L) = \nabla_{\dot{\mathbf{q}}\dot{\mathbf{q}}} L = \frac{\partial^2 L}{\partial \dot{\mathbf{q}} \partial \dot{\mathbf{q}}}. \tag{13}$$

The inverse matrix exists by the non-degeneracy condition (2). It is convenient to keep the system in the form which is linear in the accelerations and does not contain the inverse matrix,

$$\mathbf{H}(L)\, \ddot{\mathbf{q}} = A(L), \quad \text{where} \quad A(L) = \frac{\partial L}{\partial \mathbf{q}} - \frac{\partial}{\partial \mathbf{q}} \frac{\partial L}{\partial \dot{\mathbf{q}}} \dot{\mathbf{q}}. \tag{14}$$

Both $\mathbf{H}(L)$ and $A(L)$ are functions of the point $(\mathbf{q}, \dot{\mathbf{q}}) \in \Omega$: the first one takes values in $\mathbb{R}^{r\times r}$, the second one takes values in $\mathbb{R}^r$. Equation (14) is the object that carries all the information about the dynamics: two Lagrangians define the same dynamic system if and only if they produce the same pair of the coefficients of this system, up to the ambiguities described below.

Lemma 1 *Let $\Omega \subset \mathbb{R}^{2r}$ be compact, let $L \in \mathbb{L}$ be twice continuously differentiable on Ω, and let $\det \mathbf{H}(L)(\mathbf{x}) \neq 0$ for every $\mathbf{x} \in \Omega$. Then there exist $a > 0$ and $b > 0$ such that for all $\mathbf{x} \in \Omega$*

$$|\det \mathbf{H}(L)(\mathbf{x})| \geq a \quad \textit{and} \quad \min_{j=1,\dots,r} \left|\lambda_j\big(\mathbf{H}(L)(\mathbf{x})\big)\right| \geq b,$$

where $\lambda_1, \dots, \lambda_r$ are the eigenvalues of the matrix $\mathbf{H}(L)(\mathbf{x})$.

Proof. The entries of $\mathbf{H}(L)$ are the second partial derivatives of L, hence they are continuous on Ω. The determinant is a polynomial in the entries of a matrix, so the function $\mathbf{x} \mapsto |\det \mathbf{H}(L)(\mathbf{x})|$ is continuous on Ω. By the extreme value theorem a continuous real-valued function on a compact set attains its minimum, so there exists $\mathbf{x}_a \in \Omega$ with

$$a := |\det \mathbf{H}(L)(\mathbf{x}_a)| = \min_{\mathbf{x}\in\Omega} |\det \mathbf{H}(L)(\mathbf{x})| .$$

By the assumption $\det \mathbf{H}(L)(\mathbf{x}_a) \neq 0$, hence $a > 0$.

The matrix $\mathbf{H}(L)(\mathbf{x})$ is symmetric, so its eigenvalues are real and depend continuously on its entries. Therefore the function $\mathbf{x} \mapsto \min_j |\lambda_j(\mathbf{H}(L)(\mathbf{x}))|$ is continuous on Ω and, by the same argument, attains its minimum b at some point $\mathbf{x}_b \in \Omega$. If b were equal to zero, the matrix $\mathbf{H}(L)(\mathbf{x}_b)$ would have a zero eigenvalue and its determinant $\prod_j \lambda_j$ would vanish, which contradicts the assumption. Hence $b > 0$. ☐

Lemma 1 states that on a compact domain the non-degeneracy of the Hessian is uniform: the system (14) is not only solvable at every point, but its condition number is bounded. This is the property that makes the numerical solution of (14) stable and justifies the constraint introduced in Section 5. The Euler-Lagrange equation is invariant with respect to the multiplication of the Lagrangian by a positive constant, therefore the scale of $\mathbf{H}(L)$ is not determined by the trajectory, and we may fix it by requiring that the eigenvalues of $\mathbf{H}(L)$ are at least 1.

Lemma 2 *The operators*

$$A(L) = \frac{\partial L}{\partial \mathbf{q}} - \frac{\partial}{\partial \mathbf{q}}\frac{\partial L}{\partial \dot{\mathbf{q}}}\dot{\mathbf{q}} \quad \textit{and} \quad \mathbf{H}(L) = \frac{\partial^2 L}{\partial \dot{\mathbf{q}}\partial \dot{\mathbf{q}}}$$

are linear in L.

Proof. Both operators are compositions of partial differentiation, which is linear, and of the multiplication by the fixed function $\dot{\mathbf{q}}$, which does not depend on L. Explicitly, for any $L_1, L_2 \in \mathbb{L}$ and any $\alpha, \beta \in \mathbb{R}$,

$$A(\alpha L_1 + \beta L_2) = \frac{\partial(\alpha L_1 + \beta L_2)}{\partial \mathbf{q}} - \frac{\partial}{\partial \mathbf{q}} \frac{\partial(\alpha L_1 + \beta L_2)}{\partial \dot{\mathbf{q}}} \dot{\mathbf{q}} =$$
$$\alpha \frac{\partial L_1}{\partial \mathbf{q}} + \beta \frac{\partial L_2}{\partial \mathbf{q}} - \alpha \frac{\partial}{\partial \mathbf{q}} \frac{\partial L_1}{\partial \dot{\mathbf{q}}} \dot{\mathbf{q}} - \beta \frac{\partial}{\partial \mathbf{q}} \frac{\partial L_2}{\partial \dot{\mathbf{q}}} \dot{\mathbf{q}} = \alpha A(L_1) + \beta A(L_2),$$

and the same computation for the second derivatives gives $\mathbf{H}(\alpha L_1 + \beta L_2) = \alpha \mathbf{H}(L_1) + \beta \mathbf{H}(L_2)$. □

Introduce the functional

$$\|L\|_{\mathbb{L}} = \big\| \big(A(L), \mathbf{H}(L)\big) \big\|_2. \tag{15}$$

Since the map $L \mapsto (A(L), \mathbf{H}(L))$ is linear by Lemma 2 and $\| \cdot \|_2$ is a norm on the space of its values, the functional (15) is absolutely homogeneous and satisfies the triangle inequality . It is a seminorm and not a norm, because its kernel

$$\mathcal{N} = \{L \in \mathbb{L} \mid \|L\|_{\mathbb{L}} = 0\}$$

is a non-trivial linear subspace of $\mathbb{L}$: it is the intersection of the kernels of the two linear operators. The quotient space $\mathbb{L}/\mathcal{N}$ is therefore normed. The following statement identifies the kernel $\mathcal{N}$ with a classical object of analytical mechanics .

Remark 1 The kernel $\mathcal{N}$ is exactly the set of the gauge terms of the Lagrangian. In classical mechanics it is a well-known fact that two Lagrangians which differ by the total time derivative of a function of the coordinates and time, $L' = L + \frac{d}{dt} F(\mathbf{q}, t)$, produce the same Euler-Lagrange equations and hence the same trajectories; see [23, §2]. For the autonomous systems considered here $F = F(\mathbf{q})$ and

$$\frac{dF(\mathbf{q})}{dt} = \nabla_{\mathbf{q}} F(\mathbf{q}) \cdot \dot{\mathbf{q}}.$$

A direct computation gives $\mathbf{H}\big(\frac{dF}{dt}\big) = \mathbf{0}$, since this function is affine in $\dot{\mathbf{q}}$, and

$$A\left(\frac{dF}{dt}\right) = \nabla_{\mathbf{q}}(\nabla_{\mathbf{q}} F \cdot \dot{\mathbf{q}}) - (\nabla_{\mathbf{q}\mathbf{q}} F)\dot{\mathbf{q}} = (\nabla_{\mathbf{q}\mathbf{q}} F)\dot{\mathbf{q}} - (\nabla_{\mathbf{q}\mathbf{q}} F)\dot{\mathbf{q}} = \mathbf{0},$$

so $\frac{dF}{dt} + c \in \mathcal{N}$ for any constant c. Conversely, let $L \in \mathcal{N}$. The condition $\mathbf{H}(L) = \mathbf{0}$ means that L is affine in the velocities, $L = \sum_{i=1}^{r} a_i(\mathbf{q}) \dot{q}_i + b(\mathbf{q})$.

Substituting this expression into $A(L)$ gives, for the j-th component,

$$A(L)_j = \sum_{i=1}^{r} \left(\frac{\partial a_i}{\partial q_j} - \frac{\partial a_j}{\partial q_i} \right) \dot{q}_i + \frac{\partial b}{\partial q_j}.$$

This expression vanishes for all $\dot{\mathbf{q}}$ if and only if b is constant and the differential form $\sum_i a_i \, dq_i$ is closed, that is, $\mathbf{a} = \nabla_{\mathbf{q}} F$ for some function F on a simply connected domain Ω. Hence $\mathcal{N} = \{\frac{d}{dt} F(\mathbf{q}) + c\}$ and the quotient $\mathbb{L}/\mathcal{N}$ is the space of the physically distinguishable Lagrangians. We stress that the seminorm (15) is not a new equivalence: it is a way to compute the classical gauge equivalence numerically from the values of the network.

The next theorem relates the equivalence induced by the seminorm to the observable quantity, the acceleration .

Theorem 1 *Let $L, L' \in \mathbb{L}$ be non-degenerate on Ω in the sense of (2), and let $\ddot{\mathbf{q}}$ and $\ddot{\mathbf{q}}'$ be the accelerations defined by the systems (14) written for L and L' respectively. Denote $\delta L = L' - L$ and $\delta \ddot{\mathbf{q}} = \ddot{\mathbf{q}}' - \ddot{\mathbf{q}}$. Then*

1. *if $\|\delta L\|_{\mathbb{L}} = 0$, then $\delta \ddot{\mathbf{q}} = \mathbf{0}$ almost everywhere on Ω;*
2. *if $\delta \ddot{\mathbf{q}} = \mathbf{0}$ almost everywhere on Ω and, in addition, $\mathbf{H}(\delta L) = \mathbf{0}$ almost everywhere on Ω, then $\|\delta L\|_{\mathbb{L}} = 0$.*

Proof. By the definition of the accelerations,

$$\mathbf{H}(L)\ddot{\mathbf{q}} = A(L), \qquad \mathbf{H}(L')\ddot{\mathbf{q}}' = A(L'). \tag{16}$$

Substitute $L' = L + \delta L$ and $\ddot{\mathbf{q}}' = \ddot{\mathbf{q}} + \delta \ddot{\mathbf{q}}$ into the second equality and expand it using the linearity of A and $\mathbf{H}$ established in Lemma 2:

$$\begin{aligned} A(L) + A(\delta L) &= \big(\mathbf{H}(L) + \mathbf{H}(\delta L)\big)\big(\ddot{\mathbf{q}} + \delta \ddot{\mathbf{q}}\big) = \\ &= \mathbf{H}(L)\ddot{\mathbf{q}} + \mathbf{H}(L)\delta\ddot{\mathbf{q}} + \mathbf{H}(\delta L)\ddot{\mathbf{q}} + \mathbf{H}(\delta L)\delta\ddot{\mathbf{q}}. \end{aligned} \tag{17}$$

Subtracting the first equality of (16) from (17) gives the identity

$$A(\delta L) = \mathbf{H}(\delta L)\ddot{\mathbf{q}} + \mathbf{H}(L)\delta\ddot{\mathbf{q}} + \mathbf{H}(\delta L)\delta\ddot{\mathbf{q}}. \tag{18}$$

Prove the first statement. If $\|\delta L\|_{\mathbb{L}} = 0$, then by the definition (15) of the seminorm $A(\delta L) = \mathbf{0}$ and $\mathbf{H}(\delta L) = \mathbf{0}$ almost everywhere on Ω. Substituting this into (18) leaves

$$\mathbf{0} = \mathbf{H}(L)\, \delta \ddot{\mathbf{q}}. \tag{19}$$

The matrix $\mathbf{H}(L)$ is non-degenerate at every point of Ω by the assumption, therefore $\delta \ddot{\mathbf{q}} = \mathbf{0}$ almost everywhere .

Prove the second statement. Let $\delta\ddot{\mathbf{q}} = \mathbf{0}$ and $\mathbf{H}(\delta L) = \mathbf{0}$ almost everywhere. Then every term in the right-hand side of (18) vanishes, hence $A(\delta L) = \mathbf{0}$ almost everywhere and, together with $\mathbf{H}(\delta L) = \mathbf{0}$, this gives $\|\delta L\|_{\mathbb{L}} = 0$. $\square$

Remark 2 The additional assumption $\mathbf{H}(\delta L) = \mathbf{0}$ in the second statement of Theorem 1 cannot be dropped. Indeed, for $L' = cL$ with a constant $c > 0$ the system (14) is multiplied by c and produces the same accelerations, $\delta\ddot{\mathbf{q}} = \mathbf{0}$, while $\|\delta L\|_{\mathbb{L}} = |c-1|\,\|L\|_{\mathbb{L}} \neq 0$. This is the scaling ambiguity of the Lagrangian. It is the reason why the constraint on the eigenvalues of $\mathbf{H}$ is imposed in Section 5: it fixes the scale of the recovered Lagrangian and makes the representation of a trajectory reproducible. Together with Remark 1, the two statements say that the seminorm (15) measures exactly the physically meaningful part of the difference between two Lagrangians: it is blind to the gauge terms and to nothing else, and it is sensitive to the overall scale, which is fixed by the normalization.

The seminorm (15) cannot be evaluated exactly, because it requires the integration of the values of the network over the phase space. It is approximated by numerical integration. Let $\Omega \subset \mathbb{R}^{2r}$ be the compact sampling domain and let μ be the Lebesgue measure on it. By the definition of the L_2-norm ,

$$\|L\|_{\mathbb{L}} = \left(\int_{\Omega} \left(\|A(L)(\mathbf{q},\dot{\mathbf{q}})\|_2^2 + \|\mathbf{H}(L)(\mathbf{q},\dot{\mathbf{q}})\|_F^2 \right) d\mu(\mathbf{q},\dot{\mathbf{q}}) \right)^{1/2}, \tag{20}$$

where $\|\cdot\|_2$ is the Euclidean norm of a vector and $\|\cdot\|_F$ is the Frobenius norm of a matrix. The integrand is continuous on the compact set Ω, hence integrable. Let $\{(\mathbf{q}_i,\dot{\mathbf{q}}_i)\}_{i=1}^N$ be a set of probe points in Ω. The integral is approximated by the quadrature

$$\|L\|_{\mathbb{L}} \approx \left(\frac{\mu(\Omega)}{N} \sum_{i=1}^{N} \left(\|A(L)(\mathbf{q}_i,\dot{\mathbf{q}}_i)\|_2^2 + \|\mathbf{H}(L)(\mathbf{q}_i,\dot{\mathbf{q}}_i)\|_F^2 \right) \right)^{1/2}. \tag{21}$$

The probe points are either the nodes of a uniform grid, in which case (21) is a Riemann sum and converges to (20) as the mesh size tends to zero, or independent samples from the uniform distribution on Ω, in which case (21) is the Monte Carlo estimate of the integral and converges almost surely as $N \to \infty$ by the strong law of large numbers, with the error of the order $O(N^{-1/2})$ independent of the dimension $2r$. The second option is used in the computational experiment, because the number of the grid nodes grows exponentially with $2r$.

Remark 3 In many cases the second summand in (20) may be omitted. Passing to the canonical coordinates by the Legendre transform makes the Hessian $\mathbf{H}$ the identity matrix, so that $\mathbf{H}(\delta L) = \mathbf{0}$ for the Lagrangians normalized in this way and it is sufficient to compare the first part of the seminorm, $\|A(L)\|_2$. This is also what makes the vectorization of Section 6 possible: the vector of the values of $A(L)$ at a fixed finite set of probe points is a projection of the Lagrangian space onto $\mathbb{R}^n$ which changes the seminorm by an arbitrarily small value as N grows.

Since we are solving a classification problem, we assign to each Lagrangian L_i its class $\mathcal{A}_k = \{L_i \mid y_i = k\}$, where y_i is the class label of the Lagrangian. By the compactness hypothesis stated at the beginning of this section, we thus have a finite family of non-intersecting closed convex sets $\mathcal{A} = \{\mathcal{A}_k\}_{k=1}^K$ in the space $\mathbb{L}$ of Lagrangians, where each set represents a class that corresponds to one type of human motion.

Theorem 2 *Let $\mathcal{A} = \{\mathcal{A}_1, \dots, \mathcal{A}_K\}$ be a finite family of non-intersecting closed convex sets in the normed space $\mathbb{L}$. Then there exists $\epsilon > 0$ such that for any transformation ϕ of the space which satisfies*

$$\|\phi(\mathcal{A}_i) - \mathcal{A}_i\| < \epsilon, \quad i = 1, \dots, K,$$

the sets of the family $\phi(\mathcal{A})$ are pairwise strongly separable.

Proof. By the first Hahn-Banach separation theorem, for every pair of indices $i \neq j$ the sets $\mathcal{A}_i$ and $\mathcal{A}_j$ are separated by a continuous linear functional a,

$$\sup_{x \in \mathcal{A}_i} \langle a, x \rangle \leq \gamma_i < \gamma_j \leq \inf_{y \in \mathcal{A}_j} \langle a, y \rangle.$$

Since the functional a is continuous, for any $\delta > 0$ there exists a neighbourhood of zero $U(0)$ such that

$$\sup_{z \in U(0)} |\langle a, z \rangle| \leq \delta.$$

For each pair of classes $i \neq j$ put

$$\epsilon_{ij} = \frac{|\gamma_i - \gamma_j|}{3}.$$

A transformation ϕ with $\|\phi(\mathcal{A}_i) - \mathcal{A}_i\| < \epsilon_{ij}$ maps $\mathcal{A}_i$ into $\mathcal{A}_i + U(0)$ with the neighbourhood $U(0)$ that corresponds to $\delta = \epsilon_{ij}$, hence

$$\sup_{x \in \mathcal{A}_i + U(0)} \langle a, x \rangle \leq \gamma_i + \epsilon_{ij} < \gamma_j - \epsilon_{ij} \leq \inf_{y \in \mathcal{A}_j + U(0)} \langle a, y \rangle,$$

where the middle inequality holds because $2\epsilon_{ij} = \frac{2}{3}|\gamma_i - \gamma_j| < |\gamma_i - \gamma_j|$. Taking $\epsilon = \min_{i \neq j} \epsilon_{ij}$, which is positive because the family is finite, gives the required strong separability of all the pairs simultaneously . ☐

This theorem allows us to study the properties of the Lagrangians on their approximations, which simplifies the problem markedly.

Based on this theorem, we can state a sufficient condition for the approximation algorithm. There must exist $\epsilon > 0$ such that for any $L \in \mathbb{L}$ the inequality $\|L - L_\epsilon\| < \epsilon$ holds, where $L_\epsilon \in \mathbb{L}_\epsilon$ is its approximation . In other words, the distance from the approximated Lagrangian to the true one has to be small: the neural network must not overfit and must approximate the Lagrangian with sufficient accuracy. This condition is fulfilled for a correctly constructed network.

Theorem 2 yields the first method of classifying the Lagrangians . Since we have introduced a seminorm in the space $\mathbb{L}$, it generates a metric on the quotient space, and the metric methods of classification are applicable. The second option is to classify the Lagrangians by the parameters of the neural network that approximates them . The following theorem gives the conditions under which this is possible .

Theorem 3 *Let $\mathcal{A}$ be a finite family of sets in a normed space $\mathbb{L}$ and let $\epsilon > 0$ be such that for any transformation ϕ of the space with $\|\phi(\mathcal{A}_i) - \mathcal{A}_i\| < \epsilon$ the sets of the family $\phi(\mathcal{A})$ are pairwise strongly separable . Let $\psi \colon \mathbb{L}_\epsilon \to \mathbb{R}^n$ be a one-to-one continuous mapping whose inverse is continuous and convex. Then the sets $\mathcal{A}_i$ are pairwise strongly separable.*

Proof. Embed the family into the space $\mathbb{R}^n$ by the mapping ψ . In it the sets $\psi\phi(\mathcal{A}_i)$ are pairwise strongly separable , so each of them is bounded by the corresponding hyperplane,

$$\sup_{x \in \psi\phi(\mathcal{A}_i)} \langle a, x \rangle \leq \gamma_i.$$

Extend each set up to its hyperplane and consider the extended sets . Each of them is convex and compact. Since ψ^{-1} is convex and continuous, the image of an extended set under ψ^{-1} is also convex and compact. Hence the preimages of the extended sets , and therefore the sets $\mathcal{A}_i$ themselves, are pairwise strongly separable. ☐

The theorem gives sufficient conditions for the compactness hypothesis to hold for the initial sets. Its conditions are, however, quite strong: they require the dependence of the network parameters on the Lagrangian to be continuous, which is not guaranteed . This method is convenient because it does not require many features, but it is not suitable for showing that the hypothesis fails for a given dataset. If the hypothesis fails , it is better to use the methods based on the metric induced by the seminorm .

4 Formulation of the classification problem for trajectories of physical systems

For a given set of trajectories we define the classification model $p(y \mid X, D)$, where $y \in \{1, \ldots, K\}$ is the class label of a trajectory. The classification quality is defined by

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} [\hat{y}_i = y_i].$$

It is, however, physically more justified to classify the dynamical systems, which in the case under consideration are identified with the Lagrangians L, rather than the realizations of these systems in the form of trajectories. A trajectory depends on the initial conditions and on the observation window, whereas the Lagrangian depends only on the mechanical system itself. Therefore the paper considers a slightly different, but in this interpretation physically equivalent, formulation of the problem.

We need to find a classification method

$$p(\hat{y} \mid X, D) = p(\hat{y} \mid L, D)\, p(L \mid X),$$

where $\{X_i\}_{i=1}^{N}$ is a set of N trajectories of dimension r and length T, D is the given training sample, $\hat{y} \in \{1, \ldots, K\}$ is the predicted class label, and $p(L \mid X)$ matches the trajectory X to the Lagrangian L. The first factor is an ordinary classification model in the space of Lagrangians, and the second one is the map recovered by the Lagrangian neural network.

In this paper we take the point estimate $p(L \mid X) = [L = \hat{L}]$, where $\hat{L}$ is the parametric estimate of the Lagrangian obtained by a Lagrangian neural network from the regression problem of Section 2. This formulation, however, also makes it possible to use a probabilistic estimator $p(L \mid X)$ for the Lagrangian L.

5 Loss function

The straightforward loss function (7) is not suitable for the problem at hand. To evaluate it one has to solve the system (14), that is, to invert the matrix $\mathbf{H}(\hat{L})$ at every point of the trajectory. This is the same difficulty that arises for the Hamiltonian networks when the continuous phase space is discretized [24]. Two effects make the optimization problem (8) ill-posed.

First, the problem is scale-invariant. If the Lagrangian $\hat{L}$ is multiplied by a constant $c > 0$, both sides of (14) are multiplied by c and the predicted accelerations do not change, so the minimizer of (8) is defined up to an arbitrary positive factor, as stated in Remark 2.

Second, this invariance is not harmless numerically. As $\|\mathbf{H}(\hat{L})\| \to 0$ the matrix of the system becomes singular, the inverse $\mathbf{H}(\hat{L})^{-1}$ is unbounded, and an arbitrarily small perturbation of the right-hand side produces an arbitrarily large error $\delta\ddot{\mathbf{q}}$ in the predicted accelerations. The optimizer exploits this: it drives the Lagrangian towards zero, where the model formally fits any data, the equilibrium point becomes unstable and the reconstructed trajectories exhibit non-physical oscillations.

To regularize the problem we impose the constraint

$$\lambda_j\left(\mathbf{H}(\hat{L})(\mathbf{x})\right) \geq 1, \quad j = 1, \dots, r, \quad \mathbf{x} \in \Omega, \tag{22}$$

on the eigenvalues of the Hessian. Lemma 1 guarantees that this constraint does not exclude the true Lagrangian: on a compact domain the eigenvalues of a non-degenerate Hessian are bounded away from zero by some $b > 0$, and by the scale invariance the Lagrangian may always be multiplied by $1/b$ so that (22) holds. Hence the constrained problem attains the same optimal value as the unconstrained one, and in addition the scale of the solution is fixed.

We also replace the residual of the accelerations by the residual of the linear system (14) itself. Instead of solving the system for $\hat{\ddot{\mathbf{q}}}$ and comparing it with the measured $\ddot{\mathbf{q}}$, we substitute the measured acceleration into the system and penalize its discrepancy,

$$\mathcal{L}_{\mathrm{res}}(\mathbf{w}) = \frac{1}{T}\sum_{t=1}^{T}\left\| A(\hat{L})(\mathbf{x}_t) - \mathbf{H}(\hat{L})(\mathbf{x}_t)\,\mathbf{y}_t \right\|_2^2, \tag{23}$$

where $\mathbf{y}_t = \ddot{\mathbf{q}}_t$ is the measured acceleration and $\hat{L} = f(\cdot \mid \mathbf{w})$. The loss (23) does not contain the inverse matrix, so its gradient with respect to $\mathbf{w}$ remains bounded near the degenerate configurations, and the whole expression is a composition of automatic differentiation operations applied to the network. By Theorem 1 an exact estimate of the Lagrangian gives $\mathcal{L}_{\mathrm{res}} = 0$, and under the constraint (22) the eigenvalues of $\mathbf{H}(\hat{L})$ are at least one, so $\mathcal{L}_{\mathrm{res}} \geq \mathcal{L}$, where $\mathcal{L}$ is the loss (7): the residual of the system dominates the residual of the accelerations, and the minimization of $\mathcal{L}_{\mathrm{res}}$ also minimizes the error of the predicted accelerations.

The hard constraint (22) corresponds to the loss function with an infinite barrier,

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{res}} + \mathcal{L}_{\mathrm{restriction}}, \quad \mathcal{L}_{\mathrm{restriction}}(\mathbf{w}) = +\infty \cdot \left[\min_{j,t} \lambda_j\left(\mathbf{H}(\hat{L})(\mathbf{x}_t)\right) < 1\right], \tag{24}$$

where $[\cdot]$ is the indicator function. In this form the penalty is neither continuous nor differentiable, so it cannot be used with a gradient optimization method. We therefore replace it by a smooth one-sided barrier. Let

$$a(z) = \mathrm{SiLU}(z) = z\,\sigma(z), \quad \sigma(z) = \left(1 + e^{-z}\right)^{-1},$$

and let $s_{\min} = \min_{z\in\mathbb{R}} a(z) \approx -0.2785$ be attained at $z^* \approx -1.2785$. The penalty actually used in the computational experiment is

$$\mathcal{L}_{\text{restriction}}(\mathbf{w}) = \frac{1}{Tr}\sum_{t=1}^{T}\sum_{j=1}^{r}\alpha\Big(a\big(\beta\big(1-\lambda_j(\mathbf{H}(\hat{L})(\mathbf{x}_t))\big)\big) - s_{\min}\Big), \tag{25}$$

where $\alpha > 0$ scales the penalty and $\beta > 0$ controls the sharpness of the barrier. The subtraction of $s_{\min}$ makes every summand non-negative. The function (25) has the following behaviour. If an eigenvalue tends to zero, the argument of a tends to $+\beta$ and above, where $a(z) \approx z$, so the penalty grows linearly with the slope $\alpha\beta$ and repels the solution from the degenerate matrices. If an eigenvalue is large, the argument tends to $-\infty$, where $a(z) \to 0$, and the penalty saturates at the constant $-\alpha s_{\min}$, so that no pressure is applied to the well-conditioned solutions. The minimum of a summand is attained at $\lambda_j = 1 - z^*/\beta$, which tends to 1 as $\beta \to \infty$; in this limit (25) recovers the hard constraint (22). The resulting loss function is

$$\mathcal{L}_{\text{total}}(\mathbf{w}) = \mathcal{L}_{\text{res}}(\mathbf{w}) + \mathcal{L}_{\text{restriction}}(\mathbf{w}), \tag{26}$$

with $\mathcal{L}_{\text{res}}$ from (23) and $\mathcal{L}_{\text{restriction}}$ from (25). Expression (26) is the objective that is minimized in Section 7; the values of the hyperparameters α and β are given there.

6 Classification method

The classification method has the following form

$$p(\hat{y}|X, D) = p(\hat{y}|L, D)p(L|X).$$

As mentioned in Section 4, the method $p(L|X)$ assigns to each trajectory an estimate of the Lagrangian $\hat{L}$ from the linear regression problem for a Lagrangian neural network.

But it is not possible to apply a vector classification method to L directly, since L is an element of the infinite-dimensional space $\mathbb{L}$. Therefore the space $\mathbb{L}$ is projected onto a finite-dimensional space in such a way that the seminorm of the vectors under consideration changes by no more than a predetermined value $\epsilon > 0$.

The seminorm $\|L\|_{\mathbb{L}}$ is approximated on a finite set of probe points by the quadrature (21). Its summands are the values of

$$A(L)\left(\mathbf{q}_i, \dot{\mathbf{q}}_i\right) \quad \text{and} \quad \mathbf{H}(L)\left(\mathbf{q}_i, \dot{\mathbf{q}}_i\right)$$

at the probe points. Collecting these values into one vector gives the required projection of the space $\mathbb{L}$ onto a finite-dimensional space ; by Remark 3 it is sufficient to keep the values of $A(L)$ only. This vector is fed to the input of the classifier. Thus we obtain the classification method

$$
\begin{tikzcd}
X \arrow[r, "LNN"] & \hat{L} = g(x|w) \arrow[rr, "L_N^i = A(L)(\mathbf{q}_i, \dot{\mathbf{q}}_i)"] & & L_N \in \mathbb{R}^N \arrow[r, "\text{classifier: } \mathbb{R}^N \to K"] & \hat{y} \\
& Q_N \sim U([-L, L]^{2r}) \arrow[ur] & & &
\end{tikzcd}
\tag{27}
$$

Before the training phase, the set of probe points

$$
Q_N = \{\mathbf{s}_i\}_{i=1}^N \sim U\big([-c, c]^{2r}\big), \quad |Q_N| = N,
$$

at which the function values are sampled , is selected once and is then kept fixed. For a trajectory X the estimate $\hat{L}$ is found by the Lagrangian neural network. Then the values of $A(\hat{L})(\mathbf{s}_i)$ are computed at the probe points $\mathbf{s}_i \in Q_N$ and are collected into the vector $L_N \in \mathbb{R}^{rN}$. This vector is fed to the input of a vector classification method. Thus the points Q_N , the size c of the cube and the number N of the points act as hyperparameters, and the trained part of the model is the classifier $C\colon L_N \to \hat{y}$. The same set Q_N is used for all the trajectories, so the vectors L_N of different trajectories are comparable componentwise and the Euclidean distance between them approximates the distance induced by the seminorm (15) .

7 Computational experiment

The aim of the experiment is to classify multivariate time series represented as trajectories of physical systems. As subtasks, it is necessary to confirm the hypothesis of compactness of classes, to confirm in practice the theory that the Lagrangian space can be normalized by the proposed seminorm and sampled by a predetermined set of points, and to construct a classifier using it.

NOT IMPLEMENTED

A benchmark against the modern multivariate time series classifiers, ROCKET [18], MiniRocket [19], MultiRocket [20], InceptionTime [14] and the ensembles reported in the recent large-scale comparison [21], on the full 12-class protocol of PAMAP2.

7.1 Dataset

The study was conducted on the Physical Activity Monitoring (PAMAP2) dataset [22]. The dataset was collected from $M = 9$ subjects, eight men and one woman, aged 27.2 ± 3.3 years. Each subject wore three inertial measurement units and a heart rate monitor. The units are placed on the wrist of the dominant arm, on the chest, and on the ankle of the dominant side. Each unit provides a three-axis accelerometer with two ranges, a three-axis gyroscope, a three-axis magnetometer and a temperature sensor, all sampled at the frequency

100 Hz, which corresponds to the time step $h = 0.01$ s. The dataset consists of two parts. The protocol part, which is used here, contains 12 activities performed by every subject: lying, sitting, standing, ironing, vacuum cleaning, ascending stairs, descending stairs, walking, Nordic walking, cycling, running, and rope jumping. Each activity lasts 2–4 minutes. The optional part contains six additional activities performed by a part of the subjects only and is not used .

Of all the sensors presented in the dataset, the gyroscopes were selected as the most suitable ones, because their readings are the angular velocities of the body segments, that is, the generalized velocities $\dot{\mathbf{q}}$ of the mechanical model of the body, whereas the accelerometer readings mix the proper acceleration with the projection of gravity and depend on the unknown orientation of the sensor. The three units give $3 \times 3 = 9$ channels, so the number of the generalized coordinates is $r = 9$ and the dimension of the phase space is $2r = 18$. The accelerometers, magnetometers, temperature and heart rate channels are discarded.

Four of the 12 protocol activities are used in the experiment, $K = 4$: standing, walking, running, and cycling. These classes are chosen because they are the pure locomotion regimes of the whole body, they are recorded for all the nine subjects, and they differ in the mechanical model rather than in the context: a non-moving state , two gait regimes at different speed, and a periodic motion with an external constraint imposed by the bicycle. All the results reported below, both the projections in Figures 4 and 5 and the accuracies in Table 1, refer to this four-class problem. The remaining eight activities of the protocol are not used, because several of them, such as ironing or vacuum cleaning, are defined by the manipulated object rather than by the dynamics of the body, and the assumption that a single autonomous Lagrangian describes them is not justified .

7.2 Data preprocessing

Some readings in the series are lost due to the communication problems of the wireless sensors . To recover the series, spline interpolation is used , which gives the accuracy quadratic in the grid size. Numerical differentiation and integration are then used to recover the accelerations $\ddot{\mathbf{q}}$ and the coordinates $\mathbf{q}$ from the measured velocities $\dot{\mathbf{q}}$. We use the central difference for differentiation ,

$$\ddot{\mathbf{q}}_k = \frac{\dot{\mathbf{q}}_{k+1} - \dot{\mathbf{q}}_{k-1}}{2h}, \quad |E(\ddot{\mathbf{q}}_k)| \leq \frac{h^2}{6} \max_t \left| \mathbf{q}^{(4)}(t) \right|,$$

and Simpson's rule, one of the Newton–Cotes formulas, for integration,

$$\mathbf{q}_k = \int_0^{t_k} \dot{\mathbf{q}}(t)\, dt \approx \frac{h}{3} \sum_{j=1}^{k-1} \Big(\dot{\mathbf{q}}_{j-1} + 4\dot{\mathbf{q}}_j + \dot{\mathbf{q}}_{j+1} \Big), \quad |E(\mathbf{q}_k)| \leq \frac{t_k - t_0}{180} h^4 \max_t \left| \mathbf{q}^{(4)}(t) \right|.$$

The continuous recording of one activity of one subject is then split into non-overlapping segments of 500 ticks, which corresponds to 5 s of motion. This length is a compromise: it covers several periods of the slowest of the four gaits, and it is short enough for the motion to be treated as a single stationary regime described by one Lagrangian. The segmentation of the four selected activities of the nine subjects yields $N = 1372$ trajectories.

The resulting trajectories are passed as objects $X_i = (\mathbf{q}, \dot{\mathbf{q}})(t)_i$ and targets $Y_i = \ddot{\mathbf{q}}(t)_i$. Both are rescaled by the maximum absolute value per channel . The neural network presented in [4] is trained on the i-th trajectory separately and an estimate of the Lagrangian $\hat{L}_i$, mapped to the i-th trajectory , is obtained. This Lagrangian defines the dynamical system and allows recovery of the acceleration $\ddot{\mathbf{q}}(t)_i$. This yields a set of objects $\hat{L}_i$ and a set of labels y_i, where y_i is the class of the trajectory X_i.

7.3 Recovery of the Lagrangians

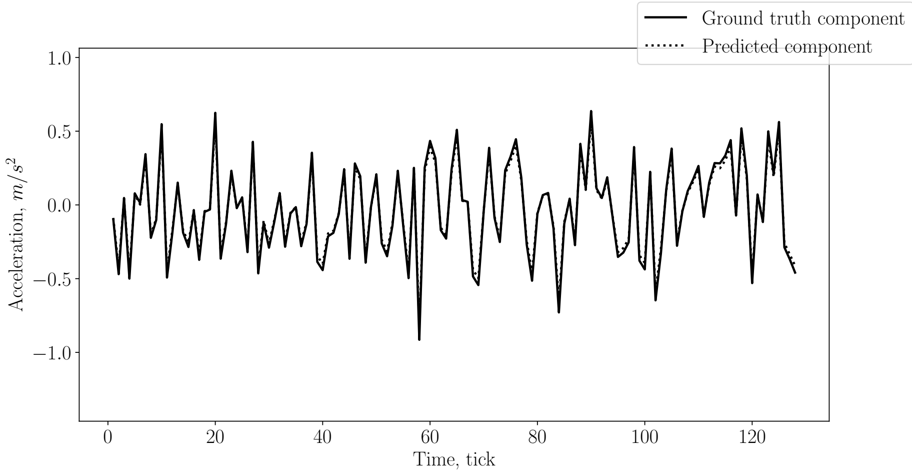
The network $\hat{L}(\cdot \mid \mathbf{w})$ maps the $2r = 18$ inputs to a scalar. It consists of an input layer $\mathbb{R}^{18} \to \mathbb{R}^{100}$ followed by the Softplus activation, two residual blocks, each of which applies Softplus and a linear layer $\mathbb{R}^{100} \to \mathbb{R}^{100}$ and adds the result to its input, and a linear output layer $\mathbb{R}^{100} \to \mathbb{R}$ with a residual connection. The Softplus activation is used because the model (11) requires the second derivatives of the output with respect to the input. The derivatives $A(\hat{L})$ and $\mathbf{H}(\hat{L})$ are computed by the forward-over-reverse automatic differentiation.

Every trajectory is fitted independently by the loss function (26) with the Adam optimizer, the learning rate 10^{-3}, the weight decay 10^{-2}, the batch size 512 and 1000 epochs. In contrast to [4], the additional regularization (25) is used. In the regions where the eigenvalues of the Hessian $\mathbf{H}$ tend to zero the equilibrium point is unstable and the reconstructed trajectories show non-physical oscillations, therefore the term (25) penalizes small eigenvalues. Its hyperparameters are adjusted depending on the type of data; for this dataset $\alpha = 5$ and $\beta = 12$ are selected, which places the minimum of the barrier at $\lambda \approx 1.11$. Fitting one trajectory takes about a minute on a single GPU, and the whole set of 1372 trajectories is processed in about a day.

As a visualization and an evaluation of the trajectory recovery with the necessary precision, the LNN predictions of the acceleration $\ddot{\mathbf{q}}(t)$ for the trajectories $(\mathbf{q}, \dot{\mathbf{q}})(t)$ were constructed and compared with the measured ones.

Figures 2, 3 show that the trajectories are exactly reconstructed by their Lagrangian, and we can predict them over a sufficiently large time interval, the losses on the test and training samples are close, and the trajectories retain their appearance. This means that, according to the assumption, the Lagrangian space is a hidden space from which all trajectories are generated.

Thus $\hat{L}_i$ vectorizes the data X_i for further classification. The seminorm on the Lagrangian space is introduced in (15), and the vectorization follows Remark 3 , in which the Lagrangian space is projected onto the Euclidean space $\mathbb{R}^n$. The vectorization of a trajectory is the vectorization of its Lagrangian .



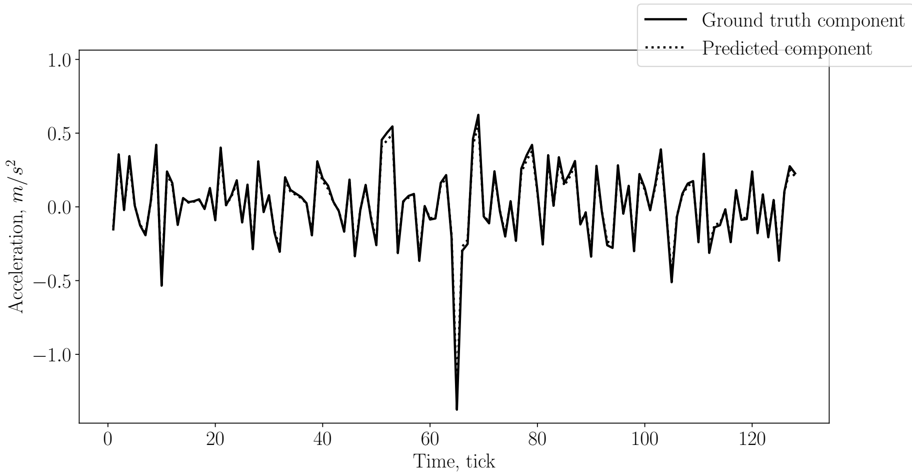


Figure 2: Time series of acceleration versus time for the test sample (a) first component, (b) second component

7.4 Vectorization and classification

For the resulting estimate of the Lagrangian, the function values can be sampled at any point $(\mathbf{q}, \dot{\mathbf{q}})$ of the phase space . Accordingly, the seminorm on the basis of which the metric is constructed is sampled, and the basic algorithms classify the trajectories with respect to this metric .

Therefore we first fix the set of points at which the values are computed . This set is chosen before the classification and is the same for all the trajectories, so it is a hyperparameter of the classifier. We draw the set $S = \{\mathbf{s}_i\}_{i=1}^N$ of $N = 20\,000$ points $(\mathbf{q}, \dot{\mathbf{q}})$ from the uniform distribution on the cube $-10 \leq x_j \leq 10$, $j = 1, \ldots, 18$. The size of the cube is chosen so that it covers

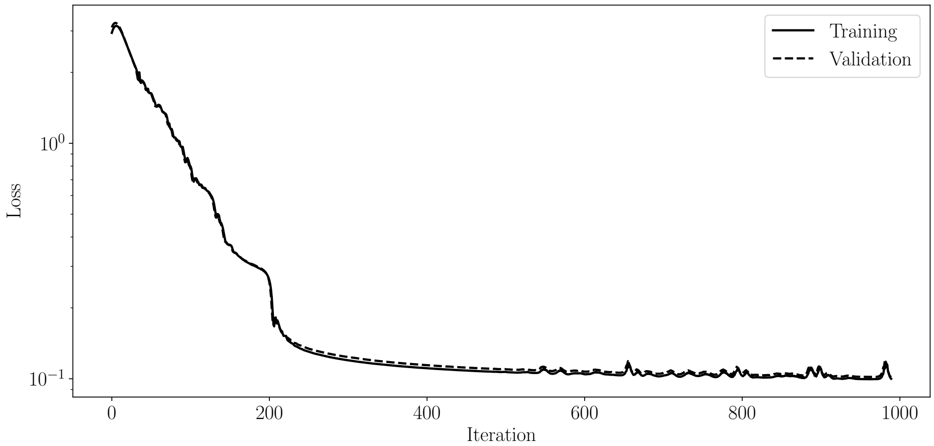


Figure 3: Model training graphics

the region visited by the rescaled trajectories on which the Lagrangians are fitted . The number of the probe points is chosen so that the estimate of the seminorm varies weakly with respect to the resampling of the set S .

For each object the features $X_i = \{A(\hat{L}_i)(\mathbf{s}_j)\}_{j=1}^N$ are obtained. Since $A(\hat{L})$ takes values in $\mathbb{R}^r$ with $r = 9$, the feature vector of one trajectory has $rN = 180\,000$ components. By (21) the seminorm in the Lagrangian space corresponds, up to the factor $\mu(\Omega)/N$, to the usual Euclidean norm in this feature space, so any vector classifier is applicable .

To regularize the model and to reduce the dimensionality, we apply the principal component analysis to the obtained features. The first two and the first three principal components are shown in Figures 4 and 5. The data are separated into clusters , one per class. The clusters visually fit the normal distribution , so we use the Gaussian kernel in the metric classifiers. The clusters are separated by linear surfaces in the presented Lagrangian space , which confirms the compactness hypothesis for this dataset .

The following classifiers are applied to the obtained representation. The random forest is chosen because it works well with complex datasets; the logistic regression and the Gaussian process classifier are chosen as simple models that are fitted quickly, so that it is useful to know whether they already solve the problem; the support vector classifier and the k nearest neighbours are the metric methods that use the induced metric directly; the Gaussian kernel is used in them because the visual analysis of Figures 4 and 5 suggests that the classes are normally distributed. The sample of 1372 trajectories is split into the training and the test parts in the proportion $80:20$ with a fixed random seed. The values in Table 1 are the means and the standard deviations over the 5-fold cross-validation on the training part.

The Gaussian process classifier shows the best results according to all the metrics in comparison to the rest of the models. The metric classifiers, the support vector

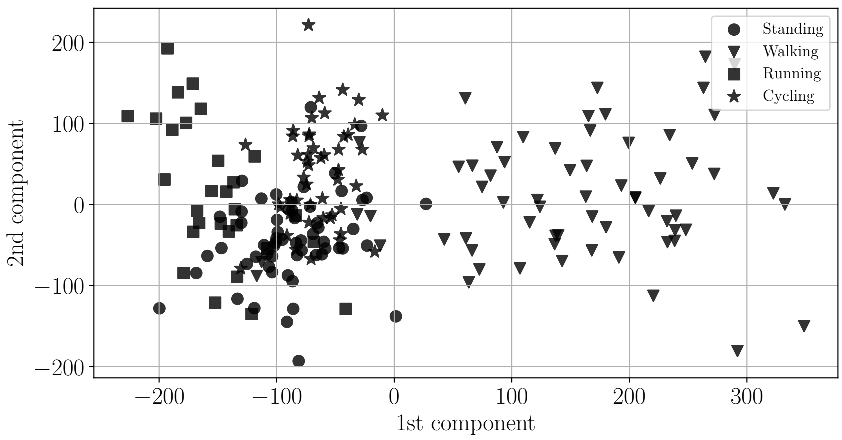


Figure 4: Projection of the Lagrangian estimates $\hat{L}$ onto the first two principal components for the four classes of the experiment: standing, walking, running, cycling

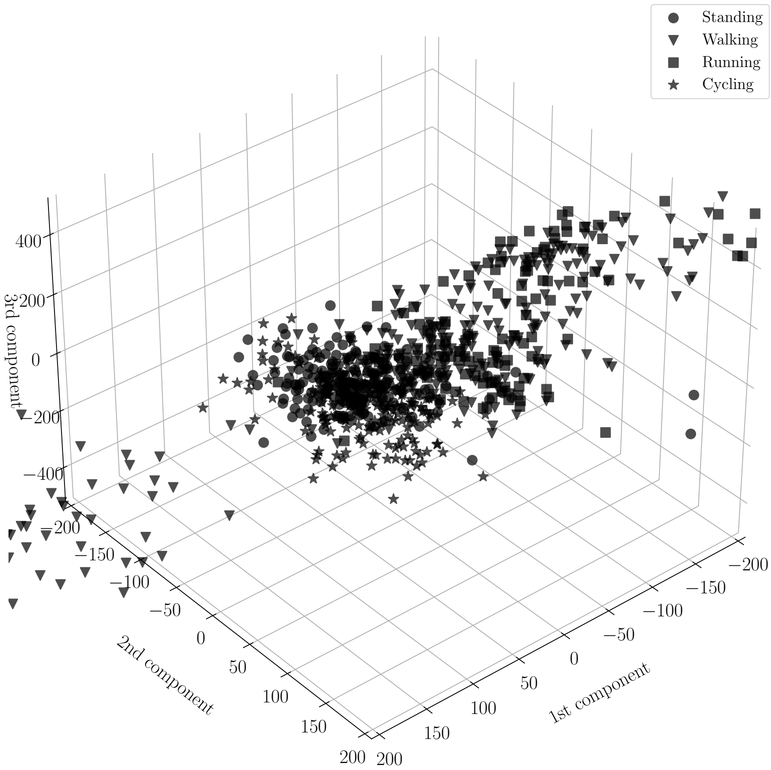


Figure 5: Projection of the Lagrangian estimates $\hat{L}$ onto the first three principal components for the same four classes

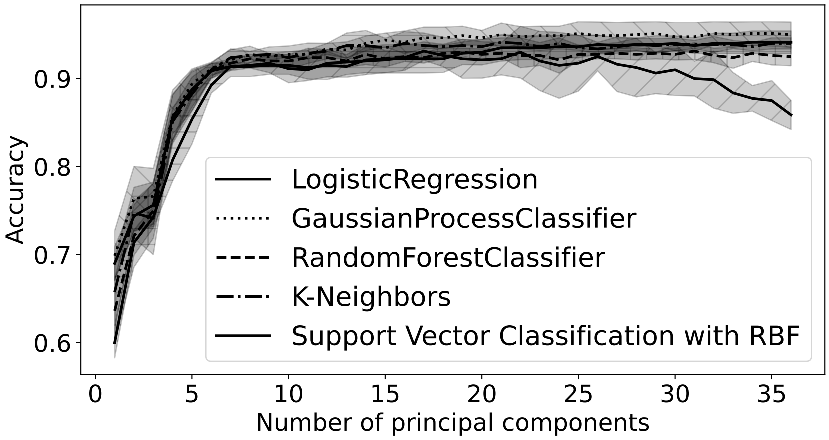


Figure 6: Classification accuracy of the selected methods depending on the number of principal components

classifier and the k nearest neighbours , show similar results. The logistic regression classifier loses in accuracy when the number of the principal components is larger than 25 , which is explained by the growth of the variance of the estimate of the separating hyperplane in the directions with a small explained variance . Among the widely-used time series classification methods, the TimeSeriesForestClassifier from the sktime library, applied to the raw gyroscope channels of the same trajectories, shows the accuracy 0.78, which is considerably worse than the accuracy of the proposed representation. Therefore we select the Gaussian process classifier.

Table 1: Accuracy of the classifiers on the proposed vectorization of the trajectories, the four-class problem, mean $\pm$ standard deviation over the 5-fold cross-validation

Classifier	Metric		
	Accuracy	Balanced-accuracy	F1-macro
Logistic regression	0.927 ± 0.014	0.924 ± 0.13	0.927 ± 0.14
Gaussian process	$\mathbf{0.946 \pm 0.010}$	$\mathbf{0.941 \pm 0.011}$	$\mathbf{0.946 \pm 0.010}$
Random forest	0.932 ± 0.007	0.928 ± 0.008	0.933 ± 0.008
K-nearest neighbors	0.939 ± 0.009	0.935 ± 0.010	0.940 ± 0.008
SVC with Gaussian kernel	0.933 ± 0.012	0.927 ± 0.013	0.933 ± 0.011

8 Conclusion

The paper proposes to classify multivariate time series of a mechanical system by the dynamic system that generates them rather than by the measurements themselves. The trajectory is mapped to the Lagrangian of the system, the

Lagrangian is recovered by a Lagrangian neural network, and the classification is performed in the space of Lagrangians.

The theoretical part of the paper introduces a seminorm on this space, built from the two coefficients of the Euler-Lagrange system (14), the operator $A(L)$ and the Hessian $\mathbf{H}(L)$. Both operators are linear in the Lagrangian, so the functional is a seminorm and the quotient by its kernel is a normed space. Remark 1 shows that this kernel is exactly the classical gauge freedom of the Lagrangian, the total time derivative of a function of the coordinates, so the construction does not introduce a new equivalence but gives a way to compute the classical one from the values of a neural network. Theorem 1 shows that the Lagrangians which are equivalent with respect to the seminorm generate the same accelerations, and that the converse holds once the scale of the Lagrangian is fixed. Theorems 2 and 3 give the conditions under which the compactness hypothesis is inherited by the finite-dimensional projection, which justifies the sampling of the seminorm at a fixed set of probe points.

The computational part shows that the construction is workable. The Lagrangian neural network reconstructs the accelerations of the PAMAP2 trajectories with the losses on the training and the test parts close to each other, Figures 2 and 3. The sampled values of $A(\hat{L})$ give a fixed-length representation of a trajectory. The method provides a convenient way to project the data onto a plane for visualization: the visual analysis of Figures 4 and 5 shows that the represented Lagrangian space corresponds to the space of features that determine the type of the motion, and the classes form compact clusters separated by linear surfaces. On the four-class problem the Gaussian process classifier reaches the accuracy 0.946, and all the five classifiers considered, including the linear one, stay above 0.92, whereas the TimeSeriesForestClassifier applied to the same raw series reaches 0.78. The representation is intrinsically multivariate: no assumptions about the relationship between the components of a trajectory are required, since all the components enter one vector of the generalized coordinates. It also preserves the physical meaning of the measurements, because every feature is the value of a differential operator applied to the reconstructed Lagrangian.

The approach has three limitations, which determine the directions of the further work.

First, and most important, the Lagrangian of every trajectory is currently recovered by a separate optimization problem. A new Lagrangian neural network is initialized and fitted for each segment, both at the training and at the test stage, so the cost of the inference on a previously unseen trajectory equals the cost of solving an ODE-constrained optimization problem, about a minute of GPU time per segment in our setting. This is acceptable for an offline study, but it is prohibitive for the real-time monitoring, which is the natural application of the activity recognition. The classifier itself is fast; the bottleneck is entirely in the map $p(L \mid X)$ from a trajectory to its Lagrangian. The natural way to remove it is to learn this map once, as an operator that takes a trajectory and returns the Lagrangian, instead of solving an optimization problem for every trajectory . The operator learning methods, DeepONet [25] and its latent-space modifications designed for the real-time prediction of the dynamics of physical systems [26], are the appropriate tools: they are trained once on a set of trajectories and then evaluated by a single forward pass, so a trajectory becomes a point in the input space of the operator rather than a separate learning task. This extension is a natural continuation of the present work, but the operator learning is outside the scope of this paper, whose subject is the structure of the Lagrangian space and the classification in it.

Second, the vectorization is sensitive to the setting of the regularization: the hyperparameters α and β of the penalty (25) have to be tuned to reach the required approximation accuracy on a particular set of trajectories, and the scale of the recovered Lagrangian is fixed only up to the choice of the bound in (22). An automatic selection of these hyperparameters, for example from the condition number of $\mathbf{H}(\hat{L})$ observed during the fitting, would make the method easier to reproduce.

Third, the empirical evaluation should be extended in two directions. The stability of the representation with respect to the non-physical perturbations of the measurements, additive noise, time warping and sensor dropout, is a plausible consequence of the fact that the representation is built from the e quations of motion rather than from the waveform, but it is a hypothesis until it is measured; the corresponding ablation study, with the perturbations applied to the test trajectories and with the augmented training sets, is left for the further work.

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